Email Address	Programme	Lecturer Name	Department	Project Title	Project Description	Which SDG does this project align with? https://sdgs.un.org/goals
fidah@um.edu.my	MAI & MDS	Prof. Ts. Dr. Rafidah Md Noor	Department of Computer System & Technology	Assessing Public Transit Reliability through Deep Learning Analytics	This project will examine delays, on-time performance, and passenger ridership trends, using data analytics to identify peak load conditions and bottlenecks, while also developing strategies to improve the overall quality of urban public transport services.	11. Sustainable cities and communities
fidah@um.edu.my	MAI & MDS	Prof. Ts. Dr. Rafidah Md Noor	Department of Computer System & Technology	Machine Learning for Pre-Trip Fatigue Detection in Bus Drivers	This project applies machine learning techniques to detect fatigue in bus drivers before trips begin. Using multimodal datasets (facial landmarks, eye closure, yawning patterns, physiological signals), predictive models will be trained to identify early signs of drowsiness, aiming to prevent fatigue-related accidents and enhance road safety.	3. Good health and well-being
rashaatallah@um.edu.my	MAI & MDS	Dr. Rasha Attaallah	Department of Computer System & Technology	Predicting Student Performance Using Machine Learning	This study applies machine learning techniques to predict student academic performance based on demographic, behavioral, and academic data. By analyzing patterns and key predictors, the research aims to identify at-risk students early. The findings can support educators in designing data-driven interventions to improve learning outcomes.	4. Quality education
rashaatallah@um.edu.my	MAI & MDS	Dr. Rasha Attaallah	Department of Computer System & Technology	AI-Driven Policy Mapping and Collaboration Network Analysis for Strengthening Partnerships in Achieving SDG 17	Global partnerships (SDG 17) are critical to achieving the Sustainable Development Goals (SDGs). However, policy documents, agreements, and collaboration networks are fragmented across nations, institutions, and organizations, making it difficult to identify synergies, overlaps, and gaps. Current analyses rely on manual review and descriptive statistics, which are inefficient and lack predictive insights. There is a need for an AI-based system that can analyze policy texts, institutional reports, and collaboration networks to provide structured insights into global SDG partnerships and policy alignments.	17. Partnership for the goals
rashaatallah@um.edu.my	MAI & MDS	Dr. Rasha Attaallah	Department of Computer System & Technology	AI-Based Fake News Detection and Disinformation Analysis for Strengthening Public Trust	This research applies natural language processing and machine learning to detect fake news and disinformation across digital media platforms. The goal is to enhance public trust by providing accurate, AI-driven insights into information credibility.	16. Peace justice and strong institutions
rashaatallah@um.edu.my	MAI & MDS	Dr. Rasha Attaallah	Department of Computer System & Technology	AI-Based Fake News Detection and Disinformation Analysis for Strengthening Public Trust	This research applies natural language processing and machine learning to detect fake news and disinformation across digital media platforms. The goal is to enhance public trust by providing accurate, AI-driven insights into information credibility.	16. Peace justice and strong institutions
rashaatallah@um.edu.my	MAI & MDS	Dr. Rasha Attaallah	Multimedia Unit	AI-Driven Student Performance Prediction and Personalized Learning Recommendation Systems	This study uses machine learning models to predict student academic performance based on demographic, behavioral, and learning data. It further develops a personalized recommendation system to provide tailored learning resources and improve educational outcomes.	4. Quality education
rashaatallah@um.edu.my	MAI & MDS	Dr. Rasha Attaallah	Department of Software Engineering	AI Models for Detecting and Reducing Algorithmic Bias in Social and Economic Decision-Making	This research investigates the use of AI models to detect and mitigate algorithmic bias in social and economic decision-making systems, such as recruitment, credit scoring, and resource allocation. By applying fairness-aware machine learning techniques, the study analyzes how biases emerge from training data and model design. The ultimate goal is to propose bias-reduction strategies that enhance transparency, equity, and trust in AI-driven decision-making.	10. Reduced inequalities
rashaatallah@um.edu.my	MAI & MDS	Dr. Rasha Attaallah	Multimedia Unit	Machine Learning for Real-Time Water Quality Prediction and Contamination Detection	Machine Learning for Real-Time Water Quality Prediction and Contamination Detection	6. Clean water

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rashaatallah@um.edu.my	MAI & MDS	Dr. Rasha Attaallah	Multimedia Unit	AI-POWERED CYBERSECURITY THREAT DETECTION IN CLOUD ENVIRONMENTS	This research leverages AI techniques to detect and mitigate cybersecurity threats in cloud environments by analyzing network traffic, user behavior, and anomaly patterns in real time. The aim is to enhance cloud security resilience, reduce attack response time, and safeguard sensitive data.	9. Industry, innovation and infrastructure
rashaatallah@um.edu.my	MAI & MDS	Dr. Rasha Attaallah	Multimedia Unit	Deepfake Detection Using AI for Digital Security	Build an AI model that detects deepfake videos and images used in cyber fraud and misinformation campaigns.	9. Industry, innovation and infrastructure
rashaatallah@um.edu.my	MAI & MDS	Dr. Rasha Attaallah	Multimedia Unit	Predictive Analytics for Heart Disease Diagnosis	Build a model based on patient data to predict heart disease.	3. Good health and well-being
rashaatallah@um.edu.my	MAI & MDS	Dr. Rasha Attaallah	Multimedia Unit	Deep Learning-Based Detection and Counting of Vehicle Number Plates in Traffic Images	This study employs deep learning techniques to automatically detect and count vehicle number plates from traffic images and video streams. The system aims to improve traffic monitoring, vehicle flow analysis, and smart city management applications.	9. Industry, innovation and infrastructure
rashaatallah@um.edu.my	MAI & MDS	Dr. Rasha Attaallah	Multimedia Unit	Predictive Analytics for Cancer Diagnosis: Build a model based on patient data to predict cancer	Predictive Analytics for Cancer Diagnosis: Build a model based on patient data to predict cancer	3. Good health and well-being
rashaatallah@um.edu.my	MAI & MDS	Dr. Rasha Attaallah	Multimedia Unit	Predictive Analytics for Stroke Diagnosis	Build a model based on patient data to predict stroke.	3. Good health and well-being
rashaatallah@um.edu.my	MAI & MDS	Dr. Rasha Attaallah	Multimedia Unit	AI-Powered Resume Screening System	Automate resume screening using NLP to match candidates with job descriptions.	9. Industry, innovation and infrastructure
rashaatallah@um.edu.my	MAI & MDS	Dr. Rasha Attaallah	Multimedia Unit	Sentiment Analysis of Customer Reviews	Objective: Build an NLP-based model to analyze and classify customer sentiment from product reviews or social media comments.	9. Industry, innovation and infrastructure
bryanraj15@um.edu.my	MAI & MDS	Dr. Bryan Raj	Department of Computer System & Technology	Predictive Analytics for Student Learning Outcomes: Leveraging Machine Learning to Enhance Inclusive and Equitable Quality Education	This project aims to develop predictive models using machine learning and educational data (e.g., student demographics, attendance, performance, and digital learning behaviors) to identify factors influencing academic success and risk of dropout. By uncovering hidden patterns, the research can help institutions personalize interventions, optimize resource allocation, and promote inclusive, equitable education for all learners.	4. Quality education
uzairiqbal@um.edu.my	MAI & MDS	Dr.Uzair Iqbal	Department of Software Engineering	Neuro-Symbolic U-NET architecture for WSI scalable dataset	This study delineates a novel neuro-symbolic deep learning framework specifically designed for whole slide imaging analysis of cancer datasets, focusing on robust generalization to unseen data. The architecture integrates symbolic reasoning with deep learning capabilities to enhance interpretability and provide explainable predictions, a critical aspect often lacking in conventional end-to-end deep learning models for medical image analysis (Syrykh et al., 2020). This comprehensive approach aims to mitigate the challenges of data heterogeneity and domain shift commonly encountered in real-world clinical applications, thereby improving diagnostic accuracy and clinical utility. The novel contribution of this neuro-symbolic reasoning approach lies in its integration with the U-Net architecture for the segmentation of WSI images, enhancing interpretability and providing explainable predictions. Furthermore, this methodology specifically addresses the inherent difficulties of generalization in deep learning models when applied to diverse cancer datasets, ensuring higher reliability across various imaging modalities and disease presentations	3. Good health and well-being

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uzairiqbal@um.edu.my	MAI & MDS	Dr.Uzair Iqbal	Department of Software Engineering	Cardiac Motion Analysis for Risk Assessment Using a Nano U-Net Architecture: Toward Lightweight and Explainable Deep Learning in Cardiovascular Imaging	Cardiovascular diseases (CVDs) are still, registering the highest number of deaths worldwide, and early and correct cardiac risk assessment is the main factor contributing to timely intervention. Traditional cardiac diagnostic methods like echocardiography, cardiac MRI, and CT provide remarkable information regarding the heart's structure and its functioning but their readings usually depend on trained professionals. Manual reading of these images is a long, subjective, and variable process. As a result, the application of advanced deep learning methods in the creation of automated, rapid, and comprehensible cardiac motion analysis systems is becoming a necessity. Among the architectures based on segmentation, U-Net has become a recognized leader in the field of biomedical image analysis. However, the traditional U-Net model is very demanding in terms of computational resources and therefore not appropriate for use in real-time clinical settings with limited resources. One solution to these problems is the Nano U-Net model, which is suggested by this research work, a lightweight but very efficient model of the classical U-Net architecture, which is specifically designed for cardiac motion analysis and risk assessment.	3. Good health and well-being

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hema@um.edu.my	MAI	Dr. Hema Subramaniam	Department of Software Engineering	An AI-Based Adaptive Learning System for Environmental Education	To develop an AI-powered educational tool that uses EEG signals to detect students' cognitive engagement during environmental science lessons, and adapts the content delivery to improve learning outcomes and climate awareness.	13. Climate action
uzairiqbal@um.edu.my	MAI	Dr.Uzair Iqbal	Department of Software Engineering	HANST: A Framework for Automated, Human-Like Clinical Reasoning	The Hierarchical Attention Neuro-Symbolic Transformer (HANST) framework is designed to automate the complex diagnostic reasoning traditionally performed by pathologists and oncologists. By holistically integrating histopathological findings from Whole Slide Images (WSIs) with data from patient records, HANST mimics expert clinical workflows without requiring direct manual input. The framework is built on four core principles: automated multimodal feature extraction, neuro-symbolic knowledge integration, transparent reasoning, and the generation of hierarchical, evidence-based explanations.	3. Good health and well-being
uzairiqbal@um.edu.my	MAI	Dr.Uzair Iqbal	Department of Software Engineering	Emergence of Neuro-Symbolic Intelligence in Federated Healthcare Learning Environments: Towards Collaborative Clinical Decision Systems	The rapid evolution of artificial intelligence (AI) in healthcare has transformed diagnostic processes, predictive analytics, and personalized treatment planning. However, traditional deep learning models, while powerful, often operate as black-box systems limiting their interpretability, trustworthiness, and clinical adoption. In contrast, Neuro-Symbolic AI (NSAI) represents a paradigm shift, integrating the perceptual capabilities of neural networks with the reasoning strength of symbolic logic. This hybrid intelligence enables systems to not only perceive and classify medical data but also reason, explain, and justify clinical decisions a critical requirement for real-world healthcare deployment. Simultaneously, the federated learning (FL) paradigm has emerged as a solution to one of healthcare's most pressing challenges: data privacy and siloed data ownership. FL enables collaborative model training across multiple healthcare institutions without sharing sensitive patient data, thereby preserving privacy while benefiting from collective intelligence	3. Good health and well-being